

□ Dacă f este o fct monotonă pe interval $[0,1]$ $f: [0,1] \rightarrow \mathbb{R}$ atunci $B_m(f)$ este o fct monotonă, de aceeași monotonie cu f .

b) Dacă f convexă sau concavă pe intervalul $[0,1]$ atunci $B_m(f)$ este convexă sau concavă pe intervalul $[0,1]$

Dem $B_m(f; x) = \sum_{k=0}^m P_{m,k}(x) f\left(\frac{k}{m}\right)$

$$P_{m,k}(x) = \binom{m}{k} x^k (1-x)^{m-k}, k=0, \overline{m}$$

a) f crescătoare $\Rightarrow B'_m(f) \geq 0$. (suficient să dem).

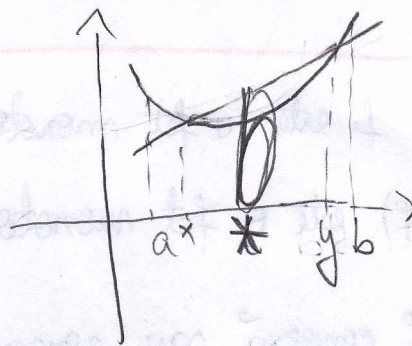
$$\begin{aligned} B'_m(f; x) &= \sum_{k=0}^m \binom{m}{k} x^k [k x^{k-1} (1-x)^{m-k} - (m-k) x^k (1-x)^{m-k-1}] f\left(\frac{k}{m}\right) \\ &= \sum_{k=1}^m \frac{m!}{(k-1)!(m-k)!} x^{k-1} (1-x)^{m-k} f\left(\frac{k}{m}\right) - \sum_{k=0}^{m-1} \frac{m!}{k!(m-k-1)!} x^k (1-x)^{m-k-1} f\left(\frac{k}{m}\right) = \\ &= m \left[\sum_{k=1}^m \binom{m-1}{k-1} x^{k-1} (1-x)^{m-k} f\left(\frac{k}{m}\right) - \sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{m-k-1} f\left(\frac{k}{m}\right) \right] \\ &= m \left[\sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{m-k-1} f\left(\frac{k+1}{m}\right) - \sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{m-k-1} f\left(\frac{k}{m}\right) \right] \\ &= m \sum_{k=0}^{m-1} P_{m-1,k}(x) [f\left(\frac{k+1}{m}\right) - f\left(\frac{k}{m}\right)] \end{aligned}$$

$$\begin{aligned} B'_m(f; x) &= m \sum_{k=0}^{m-1} P_{m-1,k}(x) \underbrace{[f\left(\frac{k+1}{m}\right) - f\left(\frac{k}{m}\right)]}_{+} = \\ &= \sum_{k=0}^{m-1} P_{m-1,k}(x) \left[\frac{k}{m}, \frac{k+1}{m}; f \right] = \sum_{k=0}^{m-1} P_{m-1,k}(x) g\left(\frac{k}{m-1}\right), \text{ , } g \in \mathcal{F} \end{aligned}$$

$$B'_m(f; x) = B_{m-1}(g; x).$$

$$g(x) = \left[\frac{x(m-1)}{m}, \frac{x(m-1)+1}{m}; f \right]$$

b) $f: J \rightarrow \mathbb{R}$, f convexa $\Leftrightarrow \forall x, y \in J, \forall \lambda \in [0, 1]: f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y)$



$$\begin{aligned} t &= (1-\lambda)x + \lambda y \\ t-1 &= \lambda(y-x) \\ x &= \frac{t-x}{y-x} \end{aligned}$$

$$f(x) - L_1(x_1, x_2; f) = (x-x_1)(x-x_2)[x_1, x_1, x_2; f]$$

Condiția necesară și suficientă pt ca o fct $f: J \rightarrow \mathbb{R}$ să fie convexă este ca pt $\forall$ 3 pt distincte $x_1, x_2, x_3 \in J$ să avem;

$$[x_1, x_2, x_3; f] \geq 0.$$

dem b:

$$B_n'(f(x)) = B_{n-1}(g, x) \quad g(x) = \left[\frac{(n-1)x}{n}, \frac{(n-1)x+1}{n}; f \right]$$

$$\begin{aligned} B_n''(f(x)) &= B_{n-1}'(g, x) = (n-1) \sum_{k=0}^{n-2} P_{n-2,k}(x) \left[g\left(\frac{k+1}{n-1}\right) - g\left(\frac{k}{n-1}\right) \right] = \\ &= \frac{(n-1)^2}{n} \sum_{k=0}^{n-2} P_{n-2,k}(x) \left(\left[\frac{k+1}{n}, \frac{k+2}{n}; f \right] - \left[\frac{k}{n}, \frac{k+1}{n}; f \right] \right) \end{aligned}$$

$$\leq \frac{2(n-1)}{n} \sum_{k=0}^{n-2} P_{n-2,k}(x) \left[\frac{k}{n}, \frac{k+1}{n}, \frac{k+2}{n}; f \right] \geq 0. \quad \text{c.c.t.d.}$$

(Scriul de polinoame Bernstein păstrează alura funcției)

Vector de poziție:

$$\vec{O} = \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$

$$(x(t), y(t), z(t)), t \in [0, 1]$$

$$\text{ex } 0, \frac{1}{n}, \dots, \frac{k}{n}, \dots, \frac{n}{n} = 1 \quad (\text{pt echidistante, val pt } t)$$

$$(B_m(x(t), t), B_m(y(t), t), B_m(z(t), t))$$

$$|f(x) - B_m(f; x)| \leq \left(1 + \frac{B_m(|t-x|; x)}{\delta}\right) \omega(f; \delta)$$

$$L_n(|t-x|; x) \leq \sqrt{L_n((t-x)^2; x)} = \sqrt{\frac{x(1-x)}{n}} = \frac{1}{2\sqrt{n}} \quad \delta = \frac{1}{\sqrt{n}}$$

$$|f(x) - B_m(f; x)| \leq \left(1 + \frac{1}{2\sqrt{n}} \sqrt{n}\right) \omega(f; \frac{1}{\sqrt{n}}) = \frac{3}{2} \omega(f; \frac{1}{\sqrt{n}})$$

$\square$ (Jackson)

Dacă f continuă $\Rightarrow$ $\exists$ sir de operatori $L_n : T \rightarrow T_m$ a.î. $f(x) - L_n(f; x) \leq c$

$$f(x) - L_n(f; x) \leq c \omega(f; \frac{1}{n}) \quad c - \text{const indep de } x \text{ și de fct. } f.$$

$\square$ Multimea polinoamelor $(P_{n,k})_{k=0}^n$ formează o bază în T_m

Dem.

$$\begin{aligned} \text{fie } P(x) &= \sum_{k=0}^m \frac{P^{(k)}(0)}{k!} x^k = \sum_{k=0}^m \frac{P^{(k)}(0)}{k!} x^k (1-x+x)^{m-k} = \\ &= \sum_{k=0}^m \frac{P^{(k)}(0)}{k!} x^k \sum_{i=0}^{m-k} \binom{m-k}{i} (1-x)^i \cdot x^{m-k-i} = \\ &= \sum_{k=0}^m \sum_{i=0}^{m-k} \frac{P^{(k)}(0)}{k!} \binom{m-k}{i} x^{m-i} (1-x)^i \stackrel{i=m-i}{=} \sum_{k=0}^m \sum_{p=k}^m \frac{P^{(k)}(0)}{k!} x^p (1-x)^{m-p} \binom{m-k}{m-p} \\ &= \sum_{k=0}^m \sum_{p=0}^m \nabla(p-k) \frac{P^{(k)}(0)}{k!} x^p (1-x)^{m-p} \binom{m-k}{m-p} = \\ &= \sum_{p=0}^m x^p (1-x)^{m-p} \sum_{k=0}^m \nabla(p-k) \frac{P^{(k)}(0)}{k!} \binom{m-k}{m-p} \\ &= \sum_{p=0}^m \frac{P_m(x)}{\binom{m}{p}} \cdot \sum_{k=0}^p \frac{P^{(k)}(0)}{k!} \binom{m-k}{m-p} \end{aligned}$$

$$P(x) = \sum_{p=0}^m a_p P_{m,p}(x) \quad , \quad a_p = \frac{1}{\binom{m}{p}} \sum_{k=0}^p \frac{P^{(k)}(0)}{k!} \binom{m-k}{m-p}$$

Algoritmul lui Zăvoianu - Castelnuovo

$$\sum_{k=0}^m a_k P_{m,k}(x)$$

folosim: $\binom{m}{k} = \binom{m-1}{k} + \binom{m-1}{k-1}$

$$\begin{aligned} \Rightarrow P_{m,k}(x) &= \binom{m-1}{k} x^k (1-x)^{m-k} + \binom{m-1}{k-1} x^k (1-x)^{m-k} \\ &= (1-x) \binom{m-1}{k} x^k (1-x)^{m-k-1} + x \binom{m-1}{k-1} x^{k-1} (1-x)^{m-k} \\ &= (1-x) P_{m-1,k}(x) + x P_{m-1,k-1}(x) \end{aligned}$$

$$P(x) = (1-x) \sum_{k=0}^m a_k P_{m-1,k}(x) + x \sum_{k=1}^m a_k P_{m-1,k-1}(x)$$

$$P(x) = (1-x) \sum_{k=0}^m a_k P_{m-1,k}(x) + x \sum_{k=0}^{m-1} a_{k+1} P_{m-1,k}(x)$$

Rezolvarea numerică a ecuațiilor neliniare

$$x^2 + 2x + 1 = 0$$

$$x^2 + 2x + 1 + \varepsilon = 0 \rightarrow \text{ec nestabilă}$$

$$\text{ex: } x^3 + x + 1 = 0$$

derivata $3x^2 + 1 \Rightarrow$ inject $\Rightarrow$ cel mult o răd.

$$\lim_{x \rightarrow -\infty} (x^3 + x + 1) = -\infty$$

$$\lim_{x \rightarrow \infty} (x^3 + x + 1) = \infty$$

$\left. \begin{array}{l} \lim_{x \rightarrow -\infty} (x^3 + x + 1) = -\infty \\ \lim_{x \rightarrow \infty} (x^3 + x + 1) = \infty \end{array} \right\} \Rightarrow$ cel puțin o răd.

$$x=0 \Rightarrow \text{neg}$$

$$x=1 \Rightarrow \text{poz}$$

$\Rightarrow$ sol în $[0,1]$

(T. pt. fix Banach)

$T: X \rightarrow X$, $\exists \alpha \in [0,1)$, $\forall x,y \in X$, $d(Tx, Ty) \leq \alpha d(x,y)$ atunci operat

T are un sg. pt fix $Tx^* = x^*$, $x_{n+1} = Tx_n$, $x_0 \in X$
converge către punctul fix $\lim_{n \rightarrow \infty} x_n = x^*$

$$d(x_n, x^*) \leq \frac{\alpha^n}{1-\alpha} d(x_1, x_0)$$

$$\boxed{ex} \quad x^3 + x - 1000 = 0$$

$$x = 1000 - x^3$$

$$x_1 \in (9, 10)$$

$$T: (9, 10) \rightarrow (9, 10) \text{ contractiv (nu-i)}$$

$$X = \frac{1000 - X}{X^2} \text{ contractiv (nu-i)}$$

$$x = \sqrt[3]{1000 - x} \text{ este contractiv}$$

$$f(x) = \sqrt[3]{1000 - x} \rightarrow f'(x) = \frac{-1}{3 \sqrt[3]{(1000 - x)^2}} \Rightarrow f' \text{ descreșc.}$$

$$\bullet f: J \rightarrow J \text{ contractiv} \quad |f'(x)| \leq \frac{1}{3 \sqrt[3]{999^2}}$$

$$|f(x_1) - f(x_2)| = |x_1 - x_2| |f'(c)| \leq |x_1 - x_2| \|f'\|$$

Def Un sir x_n are ord de convergență ρ către α dacă

$$\lim_{n \rightarrow \infty} x_n = \alpha \text{ dacă } \exists c > 0, a.î. |x_{n+1} - \alpha| \leq c |x_n - \alpha|$$

$$\rho = 1 \Rightarrow |x_{n+1} - \alpha| \leq c |x_n - \alpha|$$

$$|x_{k+1} - \alpha| \leq c |x_k - \alpha|$$

$$\prod_{k=0}^{n-1} |x_{k+1} - \alpha| \leq c^n \prod_{k=0}^{n-1} |x_k - \alpha|$$

$$|x_n - \alpha| \leq c^n |x_0 - \alpha|$$

Metoda lui Newton

$$x_n \xrightarrow[\text{către}]{\text{se convergă}} f(x) = 0$$

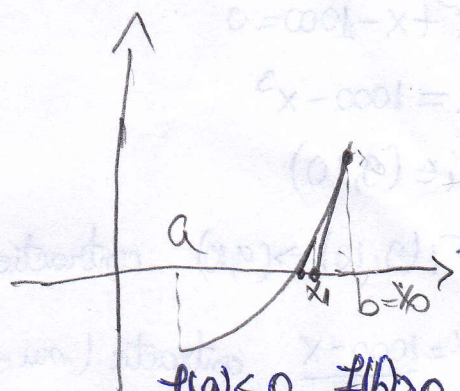
$$f: J \rightarrow \mathbb{R}, f \in C^2(J)$$

$$f(x) = f(x_n) + (x_{n+1} - x_n) f'(x_n) + \frac{(x_n - x)^2}{2} f''(\xi_n)$$

$$pp. x \approx \text{rădăcina} \Rightarrow 0 = f(x_n) + (x_{n+1} - x_n) f'(x_n)$$

$$x = x_n - \frac{f(x_n)}{f'(x_n)}, \quad f'(x) \neq 0$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad x_0 \in J$$



$$f(a) < 0, \quad f(b) > 0$$

$$f'(x) > 0, \quad f''(x) \geq 0$$

$$y = f(b) = (x-b)f'(b)$$

$$-f(x_0) = (x_1 - x_0)f'(x_0)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$\boxed{+}$ $f: [a, b] \rightarrow \mathbb{R}$ este o fct care verifică

$$f'(x) \geq 0$$

$f''(x) \geq 0$. Dacă $x_0 = b$ atunci metoda lui Newton are ordinul de convergență 2, dacă x_0 e suficient de aproape de rădăcină

dem. $f(x) = f(x_n) + (x - x_n)f'(x_n) + \frac{(x - x_n)^2}{2} f''(\xi_n)$

α -răd

$$\Rightarrow 0 = f(x_n) + (\alpha - x_n)f'(x_n) + \frac{(\alpha - x_n)^2}{2} f''(\xi_n)$$

$$0 = \frac{f(x_n)}{f'(x_n)} + \alpha - x_n + \frac{(\alpha - x_n)^2}{2} \frac{f''(\xi_n)}{f'(x_n)}$$

$$0 \leq \alpha - x_{n+1}$$

$$0 = \alpha - x_{n+1} + \frac{(\alpha - x_n)^2}{2} \frac{f''(\xi_n)}{f'(x_n)}$$

$$\frac{\alpha - x_{n+1}}{(\alpha - x_n)^2} = - \frac{f''(\xi_n)}{2f'(x_n)}$$

$$\text{fie } M = \|f''\|$$

$$m = \min f'(x), \quad x \in J$$

$$\Rightarrow \frac{|\alpha - x_{n+1}|}{|\alpha - x_n|^2} \leq \frac{M}{2m} \Rightarrow \left(\frac{M}{2m} \right)^c < 1 \Rightarrow$$

$$\Rightarrow |\alpha - x_{n+1}| \leq (\alpha - x_n)^2 c //$$